Class	Adm No

Candidate Name: ______________________________

H

millennia institute

2013 Preliminary Examination 2
Pre-University 3

Biology Higher 2 **Syllabus 9648/01**

25 September 2013

1 hr 15 min

Additional material: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

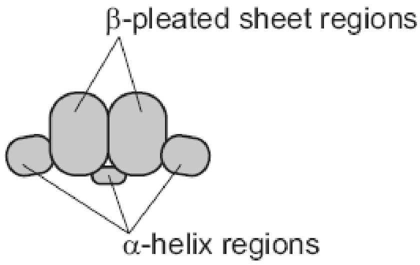
Calculators may be used.

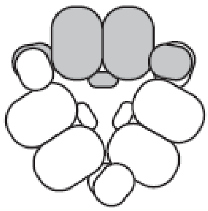
This question paper consists of 22 printed pages.

Answer **all** questions.

1. Vicilin is the major protein in pea.

- Each molecule of vicilin is made up of three identical polypeptides.
- Each polypeptide is made up of two β-pleated sheet regions with linking α-helix regions, folded into the shape shown to the right.
- This allows the three polypeptides to pack together into a compact, flat storage molecule, as shown below.





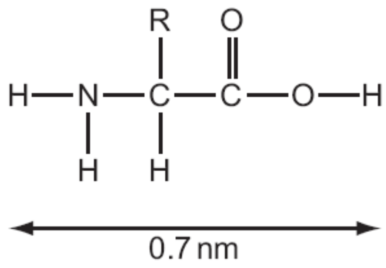
Which row correctly describes the structure of vicilin?

	primary structure	secondary structure	tertiary structure	quaternary structure
A.	amino acid sequence of one polypeptide	α-helix and β-pleated sheet regions of each polypeptide	association of three polypeptides	folding of each polypeptide
B.	amino acid sequence of one polypeptide	α-helix and β-pleated sheet regions of each polypeptide	folding of each polypeptide	association of three polypeptides
C.	association of three polypeptides	amino acid sequence of one polypeptide	α-helix and β-pleated sheet regions of each polypeptide	folding of each polypeptide
D.	association of three polypeptides	amino acid sequence of one polypeptide	folding of each polypeptide	α-helix and β-pleated sheet regions of each polypeptide

2. The diameters of some atoms when they form bonds are given in the table.

Atom	Single bond / nm	Double bond / nm
H	0.060	-
O	0.132	0.110
N	0.140	0.120
C	0.154	0.134

The approximate length of the amino acid shown below was estimated using the figures in the table.



What would be the approximate length of a dipeptide formed using this amino acid?

A. 0.8 nm

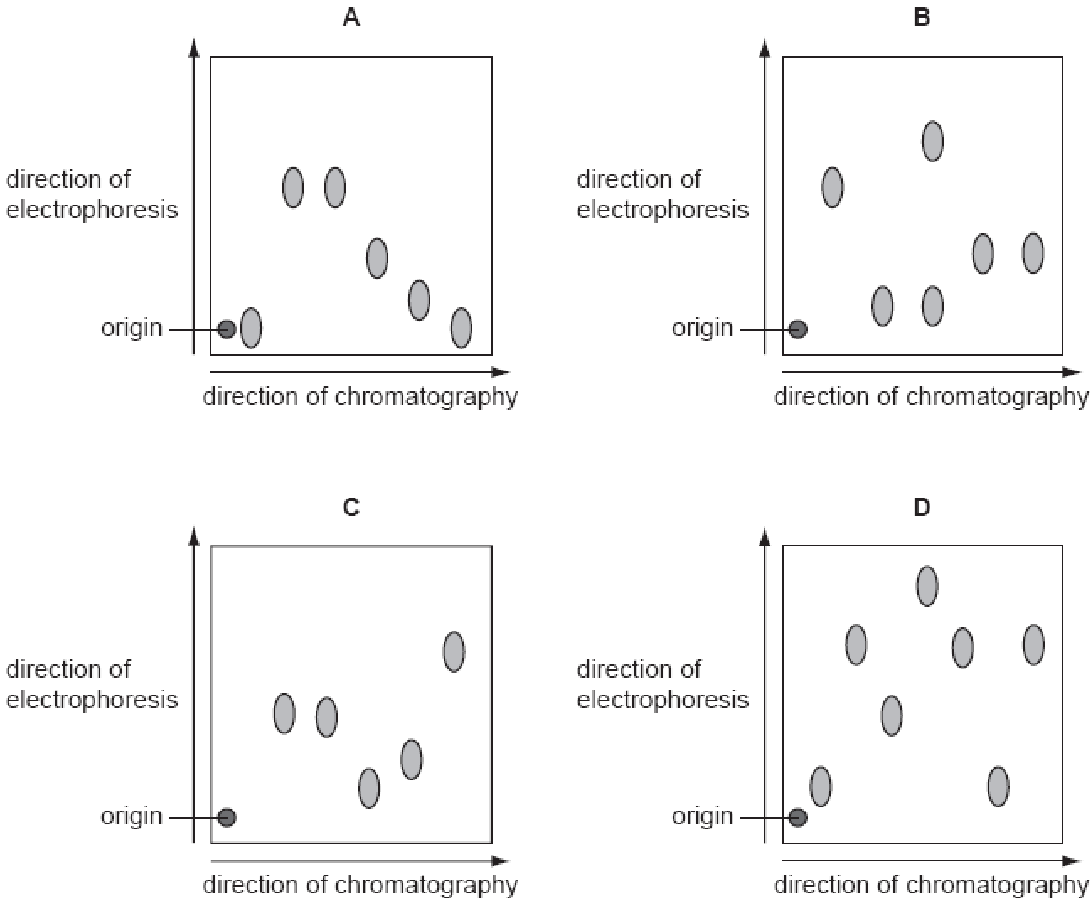
B. 1.2 nm

C. 1.5 nm

D. 1.9 nm

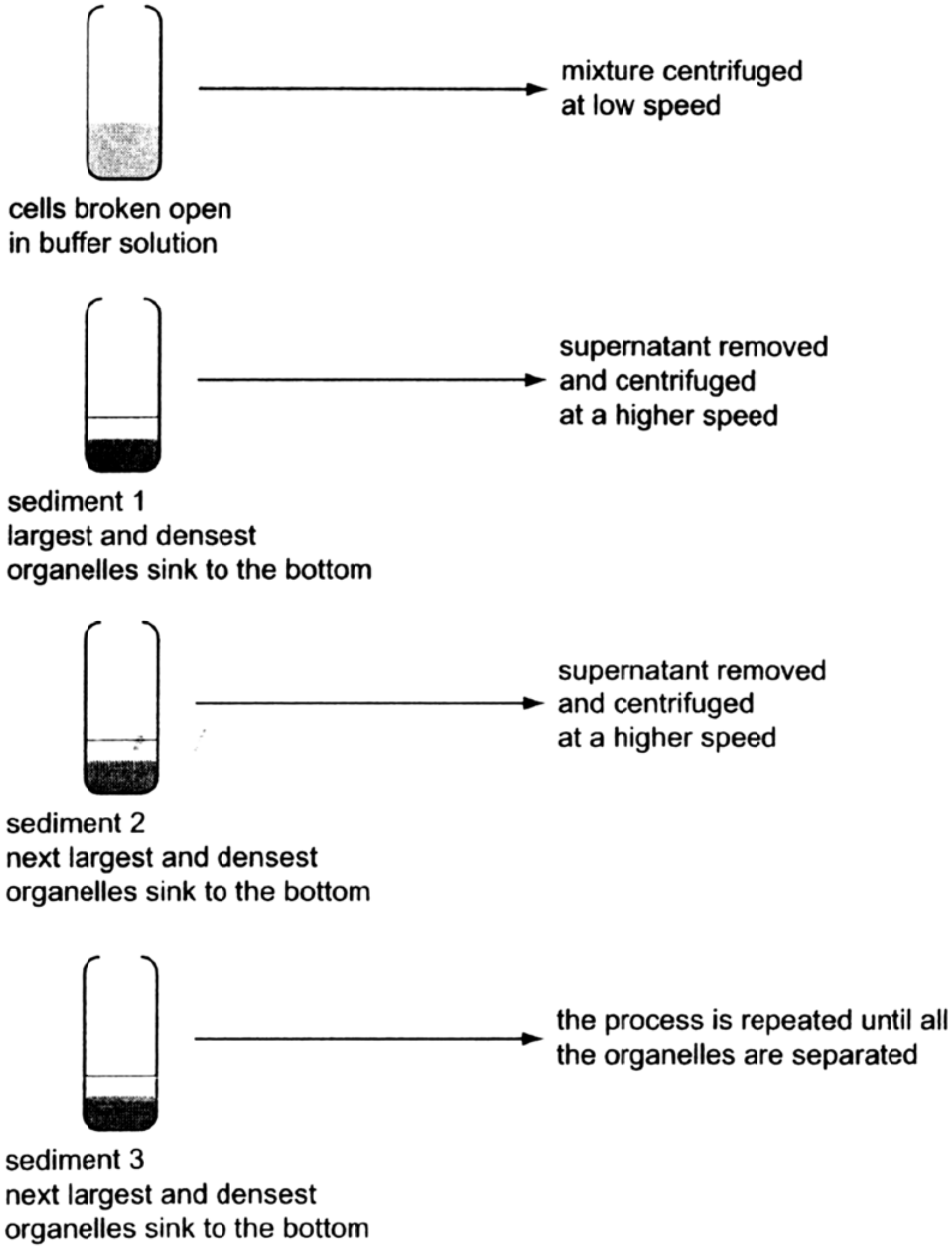
3. The diagrams show the results of an investigation into the composition of different mixtures of amino acids. Each mixture of amino acids was separated using chromatography. Each chromatogram was then turned through 90° and the amino acids separated again by electrophoresis.

Which diagram shows an amino acid mixture in which the solubility of some of the amino acids is the same but the charge on those particular amino acids is different?



4. Fractionation is used to separate plant cell components according to their size and density.

The diagram shows the main steps in fractionation of a plant cell.



DCPIP and buffer solution were added to each sediment and the mixtures left in the light for fifteen minutes. Sediment 2 caused the oxidised DCPIP to be reduced.

Which organelles are present in sediment 2?

A. nuclei

B. mitochondria

C. chloroplasts

D. ribosomes

5. Nocodazole is a chemical used in the study of mitosis. It causes all mitotic cells to stop dividing at metaphase.

Which statements correctly identify how this chemical might work?

1 inhibits chromatin condensing in the nucleus
2 prevents replication of the centrioles
3 stops sister chromatids migrating to opposite poles

A. 3 only

B. 1, 2 and 3

C. 1 and 2 only

D. 1 and 3 only

6. What is the **maximum number** of hydrogen bonds in a length of DNA containing 600 base pairs?

A. 300

B. 600

C. 1200

D. 1800

7. Which correctly describes the structure and role of a type of nucleic acid?

A. double stranded, partly base paired, unpaired bases to bind to a ribosome

B. double stranded, base sequence acts as a template for DNA synthesis

C. single stranded, partly base paired, binds to tRNA

D. single stranded, base sequence acts as a template for RNA synthesis

8. In order to replicate, the ends of a eukaryotic chromosome contain a special sequence of DNA called a telomere. Human telomeres consist of repeating TTAGGG sequences which extend from the ends of the chromosomal DNA.

When cells undergo mitotic division, some of these repeating sequences are lost. This results in a shortening of the telomeric DNA.

What is a consequence of the loss of repeating DNA sequences from the telomeres?

A. The cell will begin the synthesis of different proteins.

B. The cell will begin to differentiate as a result of the altered DNA.

C. The number of mitotic divisions the cell can make will be limited.

D. The production of mRNA will be reduced.

9. The table gives tRNA anticodons for four amino acids.

amino acid	**tRNA anticodon**
asparagine	UUA
glutamic acid	CUU
proline	GGA
threonine	UGG

A cell makes a polypeptide with the amino acid sequence:

glutamic acid – asparagine – threonine – proline

What was the sequence of bases on the strand of the DNA which was complementary to the mRNA from which this polypeptide was formed?

A. CTTTTATGGGGA

B. CUUUUAUGGGGA

C. GAAAATACCCCT

D. GAAAAUACCCCU

10. Which statements about the genetic code are correct?

1 The genetic code has redundancy and is degenerate.
2 There is only one codon for the amino acid methionine.
3 Prokaryotes generally use the same genetic code as eukaryotes.
4 Codons act as ‘stop’ and ‘start’ signals during transcription and translation.
5 mRNA codons have the same nucleotide sequence as DNA triplet codes.

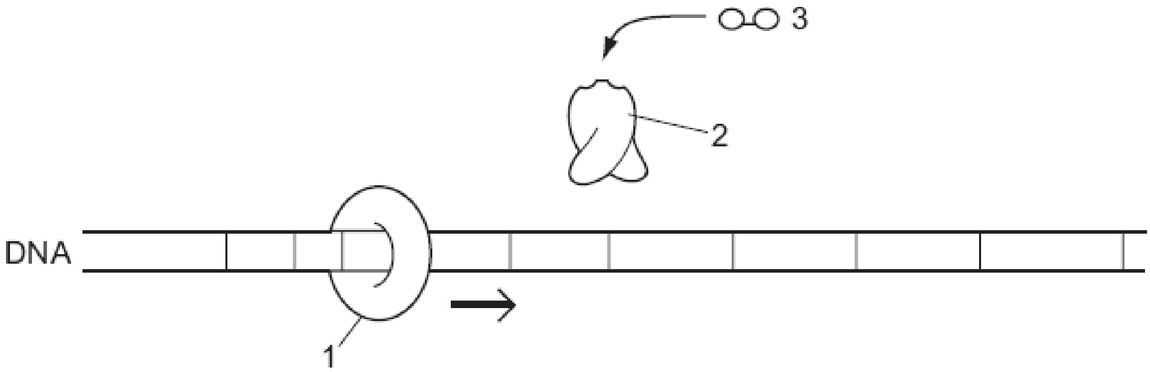
A. 1, 2 and 3

B. 1, 2 and 4

C. 1, 4 and 5

D. 2 and 4 only

11. The diagram shows the major components of the lac operon.



Which statement is correct?

A. 1 is a ribosome, 2 is a t-RNA molecule and 3 is a phosphorylated amino acid, the activator, so lactose-digesting enzyme can be made.

B. 1 is mRNA polymerase, 2 is β-galactosidase, the inducer and 3 is the repressor, so lactose-digesting enzyme cannot be made.

C. 1 is the repressor, 2 is a β-galactosidase molecule and 3 is lactose, the promoter, so lactose-digesting enzyme can be made.

D. 1 is mRNA polymerase, 2 is the repressor and 3 is lactose, the inducer, so lactose digesting enzyme can be made.

12. Which type of control of gene expression is commonly found in both prokaryotes and eukaryotes?

A. transcriptional

B. post translational

C. post transcriptional

D. translational

13. What is the normal function of tumour suppressor genes?

A. to change the coding sequences in proto-oncogenes so that they become oncogenes

B. to encode proteins that stimulate cell proliferation

C. to prevent the stimulating activity of cellular oncogenes

D. to prevent replication of cell's DNA

14. What is an example of translational control of gene expression?

A. activation of proteins by folding or enzymatic cleavage

B. addition of chemical groups, such as phosphate groups, to free amino acids in the cytoplasm

C. binding of protein factors to specific sequences in mRNA preventing ribosomes attaching

D. formation of disulphide bridges in the protein being formed

15. Which type of enzyme enables a retrovirus to prepare its genetic information for insertion into a host cell's DNA?

A. DNA polymerase

B. integrase

C. restriction enzyme

D. reverse transcriptase

16. A symbiont may be defined as a species in which individuals live in a long-term, intimate and beneficial relationship with hosts of a different species. As the name suggests, endosymbionts live within their hosts.

Which statement best provides evidence that mitochondria and chloroplasts are endosymbionts?

A. Proteins encoded by the nucleus are exported to these organelles.

B. Their inner membrane has different structure from other intracellular membranes.

C. They are surrounded by more than one membrane.

D. They contain their own ribosomes.

17. Gene mutations in either the *BRCA1* or the *BRCA2* genes are responsible for the majority of hereditary breast cancer in humans.

The proteins produced by the two genes migrate to the nucleus where they interact with other proteins, such as those produced by the tumour suppressor gene, *p53* and the DNA repair gene, *RAD51*.

Which combination of gene activity is most likely to result in breast cancer?

Key: ✓ gene produces normal protein
X gene produces abnormal protein or no protein

	gene		
	BRCA1* or *BRCA2	***p53***	***RAD51***
A.	✓	✓	✓
B.	✓	✓	X
C.	✓	X	✓
D.	X	X	X

18. Human immunodeficiency virus (HIV) is a retrovirus. After infecting a host cell, viral DNA is produced which is incorporated into the DNA of the host cell. The modified host genome now codes for the production of new HIV particles.

Which could be used as a potential treatment to slow down the spread of HIV?

1 inhibitors of restriction endonucleases
2 inhibitors of reverse transcriptase
3 restriction endonucleases
4 reverse transcriptase

A. 1 and 4 only

B. 2 and 3 only

C. 1 only

D. 2 only

19. One hypothesis about the origin of viruses is that they existed before cellular life forms, later evolving into parasites of cellular organisms. Groups with an ancient, common origin tend to share conserved sequences of DNA or RNA.

Which observation about virus genomes supports the view that viruses existed before cellular life forms and only much later evolved into parasites?

A. All viruses have genes that code for capsid components.

B. Introns of eukaryotic genes have common features with viral genomes.

C. Large virus genomes have genes that originate in host cells.

D. Viral genomes have conserved genes not found in any other genomes

20. The diagram shows an investigation into bacterial genetics

To grow mutant bacteria X
- need methionine (met^-) and biotin (bio^-)
- do not need threonine (thr^+) and leucine (leu^+)

To grow mutant bacteria Y
- do not need methionine (met^+) and biotin (bio^+)
- need threonine (thr^-) and leucine (leu^-)

↓

Grow together in the presence of methionine, biotin, threonine and leucine

↓

Grow together without methionine, biotin, threonine and leucine

↓

Growth of some bacteria that are met^+, bio^+, thr^+, leu^+

Which process or processes could explain these results?

1 transformation
2 transduction
3 conjugation

A. 1 only

B. 3 only

C. 1 and 2

D. 1 and 3

21. What is necessary for an organism to respond to changes in its internal environment?

A. a communication system between receptors and effectors

B. a communication system involving nerve cells

C. negative feedback from receptor to effector

D. positive feedback from effector to receptor

22. A snake venom causes death by leading to paralysis of muscles. It exerts its effect at synapses.

The statements below were put forward by scientists as possible explanations for the effects of this venom.

1 It interferes with the binding of neurotransmitter vesicles to the membranes.
2 It binds with neurotransmitter receptor sites.
3 It blocks calcium and sodium channels.
4 It destroys the myelin sheath of the neurone.
5 It binds with neurotransmitter.

Which statements should be investigated further?

A. 1, 2, 3 and 5 only

B. 2, 4 and 5 only

C. 1 and 4 only

D. 1, 2, 3, 4 and 5

23. *Vibrio cholerae* produces a toxin that binds to a cell surface membrane receptor on intestinal cells of the host. The toxin permanently activates the G protein in target cells, causing them to lose water rapidly. When a person is infected with cholera they suffer severe dehydration.

What statement best describes the *V. cholerae* toxin

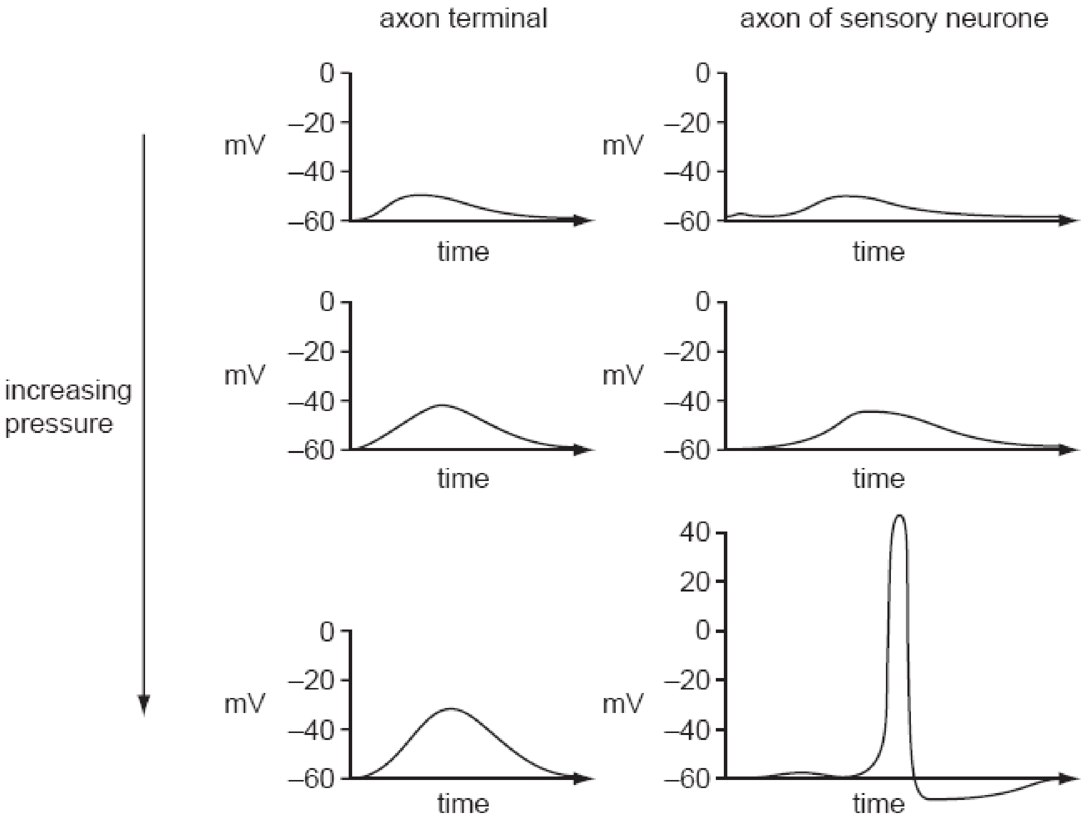
A. It is an example of a second messenger molecule.

B. It disrupts normal signal transduction in the cell.

C. It is a lipid-soluble molecule.

D. It acts as a neurotransmitter.

24. The graphs show the response of sensory neurones to increasing pressure on the Pacinian corpuscles (receptors found in deep layers of the skin that senses vibratory pressure and touch.) in mammalian skin.



Which conclusion about increasing pressure is correct?

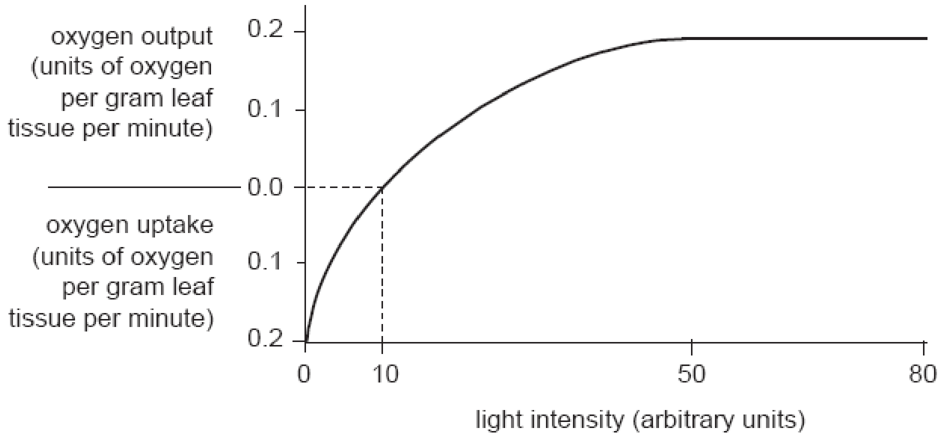
A. As pressure increases the depolarisation of the corpuscle and sensory neurone increases.

B. As pressure increases the axon terminal depolarisation reaches a value that depolarises the sensory neurone.

C. If the corpuscle is stimulated for long enough the depolarisation of the sensory neurone increases.

D. If the corpuscle is stimulated for long enough the axon terminal depolarisation reaches a value that depolarises the sensory neurone.

25. The following graph shows the relationship between light intensity and net oxygen uptake or output by a particular green plant.



Which of the following statement is true?

A. At a light intensity of 10 units, the rate of photosynthesis is zero.

B. At a light intensity of 10 units, the rate of aerobic respiration is zero.

C. At a light intensity of 10 units, oxygen produced by photosynthesis is equal to the oxygen used by aerobic respiration.

D. At a light intensity of 10 units, oxygen produced by photosynthesis is equal to twice the oxygen used by aerobic respiration.

26. What causes phenotypic variation among organisms of the same genotype?

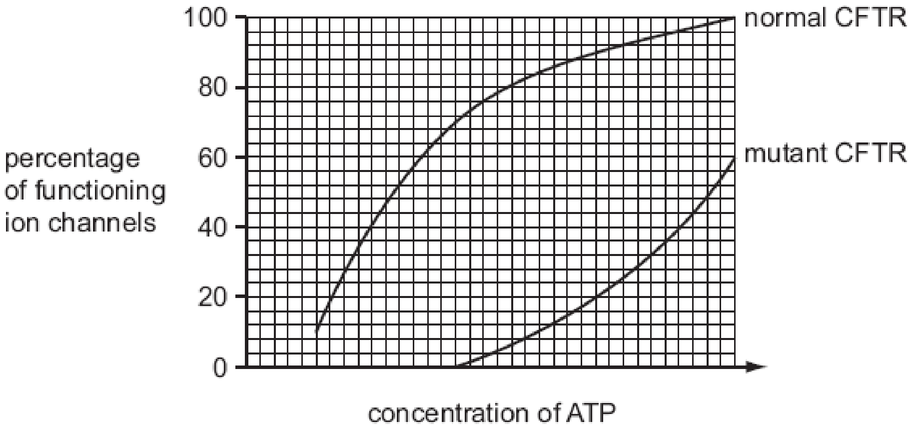
A. continuous variation within the species

B. different varieties of the same species

C. living in different environments

D. mutation

27. One of the many recessive mutations of the CFTR gene changes one amino acid in the region of the CFTR protein that binds ATP. The graph shows the effect of different concentrations of ATP on normal and mutant CFTR proteins.



Which correctly describes individuals who are homozygous for this mutation?

1 Their CFTR protein cannot bind ATP and cannot act as an ion channel.
2 Their CFTR protein binds ATP less readily than normal CFTR protein.
3 They produce CFTR protein that must bind ATP to function as an ion channel.
4 They produce a mixture of normal and mutant CFTR protein, both of which can act as an ion channel.

A. 1 only

B. 2 only

C. 2 and 3 only

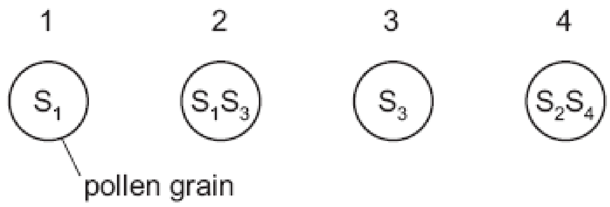
D. 2 and 4 only

28. Many plants are not fertilised by pollen from their own flowers. This is known as self-incompatibility. In any individual species, a single gene, the S gene, is responsible and it may have many different alleles.

If a pollen grain has an S allele which matches an allele in the genotype of the stigma then the pollen grain fails to germinate or the pollen tube fails to grow through the style.

The genotype of the stigma of a flower is S_3S_4.

Which pollen grains would germinate?



A. 4 only

B. 2 and 4 only

C. 1 only

D. 3 and 4 only

29. Two areas of molecular biology that have received considerable attention in evolutionary studies are the genetic code and cytochrome C. Cytochrome C is an essential component of all respiratory electron transport chains.

Which statements lend evidence to the ideas that

- all living organisms are related
- there is a single, rather than a multiple, origin of life?

1 The almost universal nature of the genetic code is a result of evolutionary convergence from multiple lineages.
2 The sequence of amino acids in cytochrome C is similar in organisms that are from similar environments or with similar metabolic demands.
3 The majority of organisms have the same, or similar, amino acid sequences for cytochrome C.
4 When transferred into a very dissimilar organism, a gene coding for cytochrome C will lead to the expression of a protein that will function in the other organism.

A. 2 and 3 only

B. 1 and 2 only

C. 3 and 4 only

D. 1, 3 and 4 only

30. The following statements relate to molecular phylogenetics.

1 Lines of descent from a common ancestor to present-day organisms have undergone similar, fixed rates of DNA mutation.

2 Organisms with similar base sequences in their DNA are closely related to each other.

3 The number of differences in the base sequences of DNA of different organisms can be used to construct evolutionary trees.

4 The proportional rate of fixation of mutations in one gene relative to the rate of fixation of mutations in other genes stays the same in any given line of descent.

Which statements, when taken together, suggest the existence of a 'molecular clock' that enables scientists to estimate the time at which one species might have diverged from another?

A. 1 and 2

B. 1 and 4

C. 2 and 3

D. 3 and 4

31. Based on binomial nomenclature, which two species are most closely related?

1 Common barberry (*Berberis vulgaris*)
2 Canadian bunchberry (*Cornus canadensis*)
3 Smooth blackberry (*Rubus canadensis*)
4 Canadian barberry (*Berberis canadensis*)

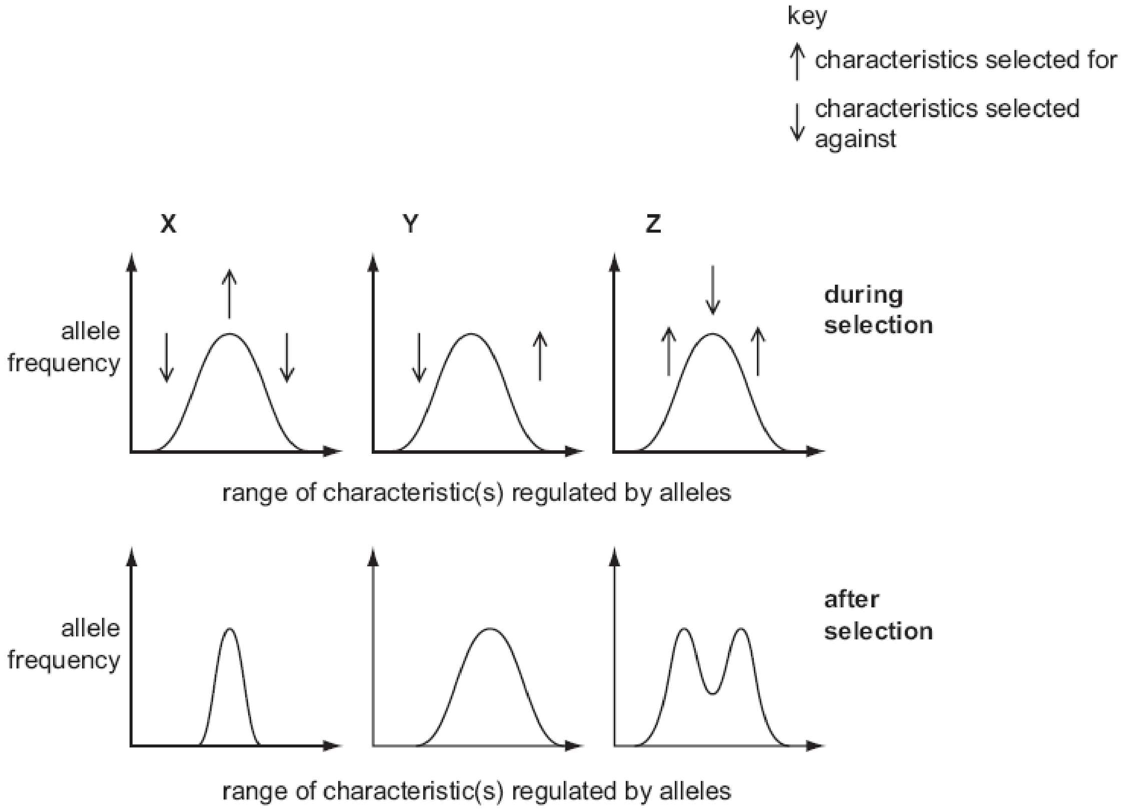
A. 1 and 4

B. 1 and 3

C. 2 and 4

D. 3 and 4

32. The graphs represent the frequency of alleles in species **X**, **Y** and **Z** during and after selection.



In which species does evolution take place?

A. **X** only

B. **Y** only

C. **Y** and **Z**

D. none of **X**, **Y** nor **Z**

33. Why can restriction fragment length polymorphism (RFLP) be used to detect inherited diseases?

A. Different alleles vary in the sequence and number of base pairs.

B. The DNA fragments are the same length within each gene.

C. The restriction site is unaffected by the number of base pairs.

D. Alleles of different genes have different molecular masses.

34. Ripening of fruit, for example figs, is caused by ethylene gas, a lipid-soluble compound that diffuses easily from cell to cell as well as to neighbouring plants. Plant tissues produce ethylene at certain developmental stages and in response to stressful conditions, such as heat or injury.

Plant cells have special ethylene-binding receptors. When ethylene is detected, a series of reactions follows inside the cell. This results in the production of pectinases to break down cell walls and amylases to convert carbohydrates into simple sugars.

Which of the following statements is consistent with the information above?

A. Ethylene stimulates cell elongation which enables fruit to grow.

B. The scratching of the surfaces of harvested figs speeds up ripening.

C. Changes triggered by ethylene would discourage animals from consuming fruit.

D. Ethylene receptors are located on the external surfaces of the cell surface membrane.

35. Stem cells are widely used in medical research.

Which property of stem cells makes them particularly useful in this research?

A. They can be fused together to form a zygote.

B. They can divide and eventually give rise to a whole organism.

C. They can divide and be made to differentiate into various types of cell.

D. They will continue to divide indefinitely.

36. Which of these processes could increase crop yield?

P inserting genes for vitamin A production into rice
Q inserting genes for pest resistance into cotton
R inserting genes for herbicide resistance into oilseed rape
S inserting genes for nitrogen fixation into soya beans

A. P and Q

B. Q and R

C. R and S

D. S and P

37. Which genetic modification could increase the yield of a crop measured as mass of crop produced per unit area per year?

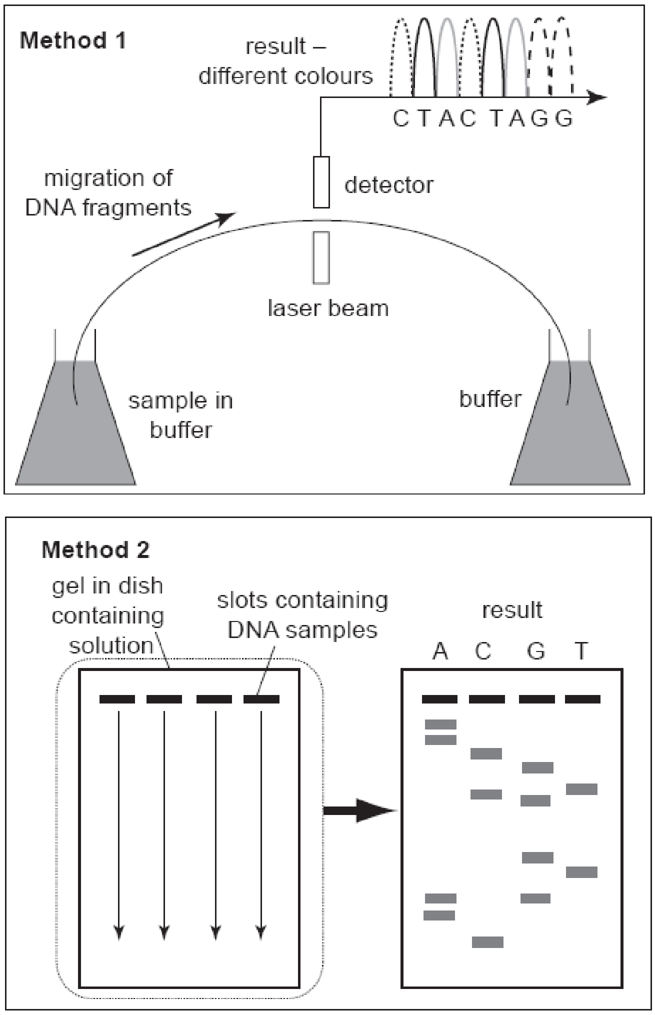
A. winter resistant plants

B. delayed ripening in the fruits

C. more essential amino acids in the seeds

D. more vitamin A in the grain

38. The following figures show two different methods used to sequence DNA.



What of the following statements is correct?

A. Method 1 requires 4 slots for the loading of the DNA.

B. Method 2 uses a laser beam to separate the individual nucleotides.

C. Method 1 uses fluorescent dyes to distinguish between the different nucleotides.

D. Method 2 requires the DNA to be loaded at the positive end on the sequencing gel.

39. Gene therapy is a way of treating genetic disease by introducing a piece of DNA into the cells of an affected individual. Liposomes can be used for gene therapy as they target the cells affected by a genetic disease.

What feature of a cell surface membrane allows the liposome to target cells affected by a genetic disease?

A. carrier molecules

B. phosphate groups

C. receptor molecules

D. protein channels

40. Small samples from crime scenes can be genetically profiled (DNA finger printed).

Which are necessary parts of a successful genetic profiling process?

Key: ✓ present
X absent

	crime scene sample	**PCR**	**ethidium bromide and X-rays**
A.	red blood cells	✓	X
B.	saliva	X	✓
C.	semen	X	✓
D.	skin cells	✓	X